ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

TECHNICAL PROGRAMMING

PAPER 2 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 2 style (design-related), built around one case study: a robot temperature logger in Python. Section A: objective questions (30) · Section B: flowcharts and a menu program, class diagrams, inheritance, polymorphism and encapsulation, saving and loading CSV files, debugging, test data and an Arduino C fault (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Program code is written in Python 3, except in QUESTION 5.3 (Arduino C). Write your code neatly, with the correct indentation.
6. Draw diagrams neatly, using a ruler.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Code that repeats until the user chooses "Quit" is a …

A if statement. B while loop.
C comment. D import statement.

1.1.2 Testing with a value exactly at the edge of a valid range is called … testing.

A normal B invalid C random D boundary

1.1.3 The method that turns "R1,55" into ["R1", "55"] is …

A split(","). B join(). C replace(). D find().

1.1.4 Using the same method name status() in different classes that each behave differently is …

A inheritance. B encapsulation.
C polymorphism. D iteration.

1.1.5 A logic error …

A stops the program from starting. B gives the wrong result without crashing.
C is always a spelling mistake. D is always found by the interpreter.

1.1.6 The function that writes a Python dictionary to a JSON file is …

A json.dump(). B json.load().
C json.read(). D json.open().

1.1.7 The three parts of a class box in a UML class diagram are …

A input, process and output. B start, loop and end.
C name, attributes and methods. D file, folder and drive.

1.1.8 A leading underscore, as in self._battery, shows that the attribute …

A is a number. B is a constant.
C has been deleted. D should only be used inside the class.

1.1.9 A tool that runs a program line by line and shows the values of variables is a …

A compiler. B debugger. C editor. D linker.

1.1.10 In Arduino C, every statement must end with a …

A colon. B full stop. C comma. D semicolon.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A flowchart shows the steps and decisions of a program using symbols.

1.2.2 Pseudocode must be written exactly in Python syntax.

1.2.3 Opening a file that does not exist for reading raises a FileNotFoundError.

1.2.4 A child class cannot add new methods of its own.

1.2.5 Comments are ignored when the program runs.

1.2.6 Test data should include invalid values to check that the program does not crash.

1.2.7 A program that runs without crashing always gives the correct results.

1.2.8 A CSV file can be opened in a spreadsheet program.

1.2.9 In Python, a single = compares two values.

1.2.10 A menu program normally uses a loop so that the menu shows again after each choice.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Diagram that uses symbols to show the flow of a program	A	encapsulation
1.3.2	Plain-language steps written before coding	B	JSON
1.3.3	A mistake in a program	C	flowchart
1.3.4	Value exactly at the limit of a valid range	D	comment
1.3.5	Checking that user input is sensible before it is used	E	validation
1.3.6	Hiding data inside a class and giving access through methods	F	bug
1.3.7	The same method call works differently for different objects	G	pseudocode
1.3.8	Text format that stores data as name–value pairs, like a dictionary	H	test plan
1.3.9	Planned list of inputs with their expected outputs	I	polymorphism
1.3.10	Note in the code that explains what it does	J	boundary value

(10)

TOTAL SECTION A: [30]

SECTION B

CASE STUDY. A warehouse uses robot arms and AGVs (automated guided vehicles). The engineers need a Python program that records the motor temperature of each robot, saves the readings to a file and warns when a motor becomes too hot. A temperature below 60 °C is OK, 60 °C to 79 °C is a warning and 80 °C or more means the robot must stop.

QUESTION 2: PROGRAM DESIGN

2.1 Draw a flowchart for the main menu of the program. The menu offers: 1 Add a reading, 2 Show all readings, 3 Show the average, 4 Quit. The menu must repeat until the user chooses 4. (6)

2.2 Write the Python menu program. Store the readings in a list. Option 1 asks for a temperature and adds it to the list; option 2 prints every reading; option 3 prints the average; option 4 ends the program; any other choice prints "Invalid choice". (8)

2.3 Explain why input validation is needed in option 1 and how it can be done. (2)

[16]

QUESTION 3: CLASSES

3.1 The parent class below is used for all the robots.

```
class Robot:
    def __init__(self, name, battery=100):
        self.name = name
        self._battery = battery

    def status(self):
        return self.name + " battery " + str(self._battery) + "%"
```

3.1.1 Draw a UML class diagram showing Robot as the parent class with TWO child classes, Arm and AGV. Show at least ONE extra attribute and ONE method for each child class. (6)

3.1.2 Write the class Arm that inherits from Robot, adds the attribute joints (number of joints) and overrides status() so that it also returns the number of joints. (6)

3.2 Explain polymorphism, and write a loop that prints the status of every robot in a list called fleet that contains Arm and AGV objects. (4)

3.3 Explain what encapsulation means and write a method get_battery() for the Robot class that returns the battery level. (4)

[20]

QUESTION 4: FILES

4.1 Write a function save_log(readings, filename) that writes each reading to a CSV file. Each reading is a tuple (time, robot, temperature), for example ("08:00", "R1", 55), and must be saved as one line: 08:00,R1,55 (5)

4.2 Write a function load_log(filename) that reads the CSV file and returns a list of tuples with the temperature as an integer. If the file does not exist, the function must return an empty list instead of crashing. (6)

4.3 The file log.csv contains the lines shown below. Write down the exact output of the code. (3)

```
08:00,R1,55
08:05,R2,63
08:10,R1,81
08:15,R3,60
```

```
count = 0
with open("log.csv") as f:
    for line in f:
        parts = line.strip().split(",")
        if int(parts[2]) >= 60:
            count = count + 1
print(count, "warnings")
```

4.4 Give TWO reasons why the readings should be saved in a file instead of only being kept in a list. (2)

[16]

QUESTION 5: TESTING AND DEBUGGING

5.1 The function below should return the average of a list of temperatures, but it contains FOUR errors. Identify EACH error and write the corrected line. (8)

```
def average(values)
    if len(values) = 0:
        return 0
    total = 0
    for v in values:
        total = total + v
    return total / len(value)

temps = [55, 62, 71]
print("Average: " + average(temps))
```

5.2 A function check_temp(t) returns "OK", "WARN" or "STOP" according to the limits in the case study. Draw up a test table with SIX test values: include normal, boundary and invalid data, and give the expected result for each. (6)

5.3 The Arduino C program below should flash an LED, but it will not compile. Identify the TWO errors and correct them. (4)

```
int ledPin = 13

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(ledPin, HIGH);
  delay(500);
  digitalwrite(ledPin, LOW);
  delay(500);
}
```

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER